

In vitro antimutagenicity of curcumin against environmental mutagens.

The effects of curcumin, the yellow pigment of the spice, turmeric (*Curcuma longa*) on the mutagenicity of several environmental mutagens were investigated in the Salmonella/microsome test with or without Aroclor 1254-induced rat-liver homogenate (S-9 mix). With *Salmonella typhimurium* strain TA98 in the presence of S-9 mix, curcumin inhibited the mutagenicity of bidi and cigarette smoke condensates, tobacco and masheri extracts, benzo[a]pyrene and dimethyl benzo[a]anthracene in a dose-dependent manner. Curcumin did not influence the mutagenicity without S-9 mix of sodium azide, monoacetylhydrazine and streptozocin in strain TA100 nor of 4-nitrophenylenediamine in strain TA98. Our observations indicate that curcumin may alter the metabolic activation and detoxification of mutagens.